Q. Travelling salesman problem

1. Problem statement: There is a list of cities and distances between every pair, what is the shortest possible route that visits every city exactly once and return to the start city

2. Objective: Cost should be minimum.

3. Pseudo code:

FUNCTION TSP {

Input: Random population of routes.
Distance (fitness) for each route is calculated
Repeat for set of generations:
a. Keep the single best route found so far
b. Loop until new population is full
- Select two parent routes (shorter)
- mutate the child route with small probability
- add the child route to the new population
c. Update the current population with new one
END and give output
Output: absolute shortest path found

}

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Q2 Genetic Algorithm for optimization problems.

Algorithm

Take an function as input for this optimization.

Algorithm:

START

1. Generate an initial population randomly
2. Evaluate its fitness that is fitness of each individual
3. Repeat until the stopping condition is satisfied
   i) Select parent individuals based on fitness
   ii) Generate offsprings
   iii) Apply mutation offspring with small probability.
   iv) Evaluate the fitness of offsprings
   v) Replace old individuals with offsprings.
4. Return the best individual as the optimal solution

END

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